

Library catalogue

Functional and Non-Functional Requirements

Functional:

- Manage books, authors, and categories.
- Filter books by title, author, category, and availability.
- Support pagination.
- User authentication.
- Return meaningful status codes.
- Follow RESTful principles (Richardson maturity model).

Non-Functional:

- Fast response time (optimized endpoints).
- Use caching where data changes rarely (e.g., categories).
- Scalable and secure architecture.
- Proper API versioning (e.g., /api/v1/...).

1. Describe what entities:

Entity	Fields
Book	id
	title
	authorId
	categoryId
	publishingYear
	availability
Author	id
	name
	birthYear
Category	id
	name

2. API functions:

Entity	Function
Book	Create book (POST localhost:8080/catalogue)
	Get book by id (GET localhost:8080/catalogue/id)
	Get all books (GET localhost:8080/catalogue)
	Update book status (PUT localhost:8080/catalogue/id?available={ture/false})

	Delete book (DELETE localhost:8080/catalogue/id)
Authors	Create author (POST localhost:8080/authors)
	Get author by id (GET localhost:8080/authors/id)
	Get all authors (GET localhost:8080/authors)
	Get all books from author (GET localhost:8080/authors/id/books)
	Delete author (DELETE localhost:8080/authors/id)
Category	Create category (POST localhost:8080/ categories)
	Get category by id (GET localhost:8080/ categories /id)
	Get all categories (GET localhost:8080/categories)
	Get all books from category (GET localhost:8080/ categories /id/books)
	Delete category (DELETE localhost:8080/ categories /id)

3. Status codes

Code	Meaning
200	OK
201	Created
204	No content
400	Bad Request
401	Unauthorized
403	Forbidden (auth but not allowed)
404	Not found
422	Unprocessable Entity
500	Internal Server Error

4. Richardson Maturity Model

Level	Feature	Included
0	Single endpoint	false (/catalogue, /authors, /categories)
1	Resources	true
2	HTTP methods	true
3	Hypermedia	false

5. Authentication

OAuth 2.0

Endpoints requiring auth: POST, PUT, DELETE

6. Caching to reduce unnecessary load on static data:

GET /categories — use Cache-Control: public, max-age=86400

GET /authors — use Cache-Control: public, max-age=86400

GET /catalogue — use Cache-Control: public, max-age=86400